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EED 1005
INTRODUCTIONS TO PROGRAMMING
BINARY SUDOKU

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OUTLINE

What is binary Sudoku

Columns and Rows

Game elements

Calculating Score

CHAPTER 1

BINARY SUDOKU

What is Binary Sudoku?

- Binary Sudoku is playing on 9X9 board.
- Goal of player is filling a column,row or 3X3 blocks and get points.
- Board will be filled with 10 different game elements that have given before.

Game's rules

- Player only put the pieces at empty point.
- If selected point is full,game take input from player again.
- A column,row or block gets filled, it will clean out and add point to score.
- New pieces generating randomly.
- If there is no suitable places for new generated pieces game over.

```
Welcome to Binary Sudoku
  A B C D E F G H I
+---+---+---+
1 | 0 . 1 | 0 . 1 | 0 . 0 |
2 | 0 1 . | . 0 1 | . . . |
3 | . . 1 | 1 . . | . . 1 |
+---+---+---+
4 | 0 . 0 | . 0 . | . 1 . |
5 | 1 . . | 0 1 1 | . . 1 |
6 | 1 . . | . 1 . | 0 1 1 |
+---+---+---+
7 | . . . | . 0 0 | 0 . . |
8 | . . . | . . . | . . . |
9 | . . . | . . . | . . . |
+---+---+---+
Score: 0

New Piece:
1 1
Enter location (ex A1): Invalid move, try again.
```

CHAPTER 2

COLUMNS AND ROWS

- Columns and rows were arranged to create 3X3 blocks.
- Game board have totally 3X3 of these blocks.

- *This function prints the current state of the board to the screen.*
- *Used to track the game flow.*
- *Players see the final state of the board.*

```
void printBoard(void) {
    printf("  A  B  C  D  E  F  G  H  I\n");
    printf("\n +-----+-----+-----+\n");
    for (int i = 0; i < SIZE; i++) {
        printf("\n%d |", i + 1);
        for (int j = 0; j < SIZE; j++) {
            if (j == 2 || j == 5 || j == 8) {
                if (board[i][j] == -1) {
                    printf(" . "); // Boş hücreyi "." olarak yaz
                } else {
                    printf(" %d |", board[i][j]);
                }
            } else {
                if (board[i][j] == -1) {
                    printf(" . ");
                } else {
                    printf(" %d ", board[i][j]);
                }
            }
        }
        if (i == 0 || i == 1 || i == 2 || i == 4 || i == 6 || i == 7) {
            printf("\n |          |          |          |");
        }
        if ((i + 1) % 3 == 0) {
            printf("\n +-----+-----+-----+\n");
        }
    }
}
```

CHAPTER 4

GAME ELEMENTS

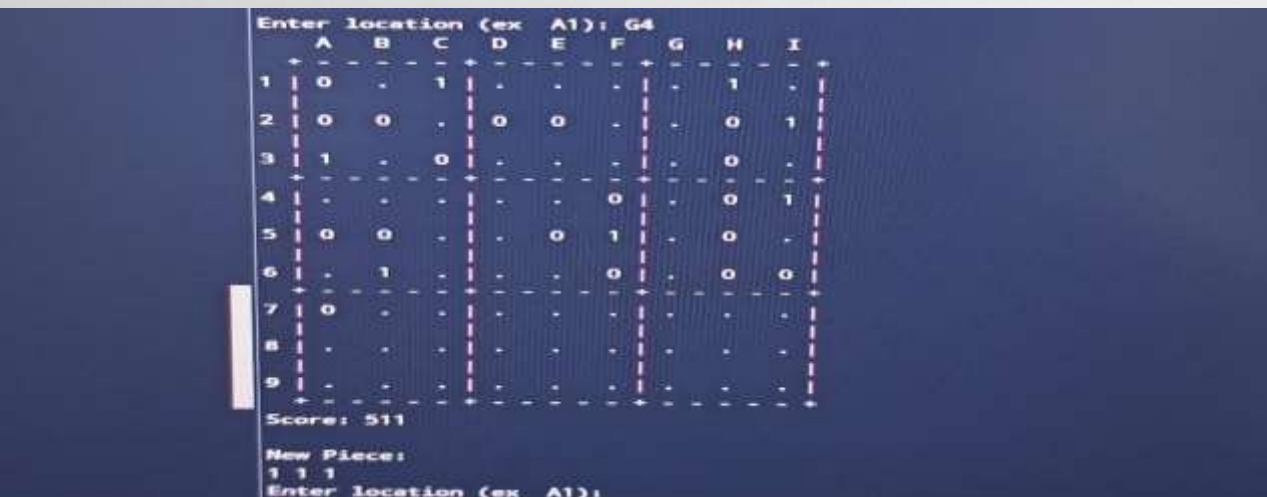
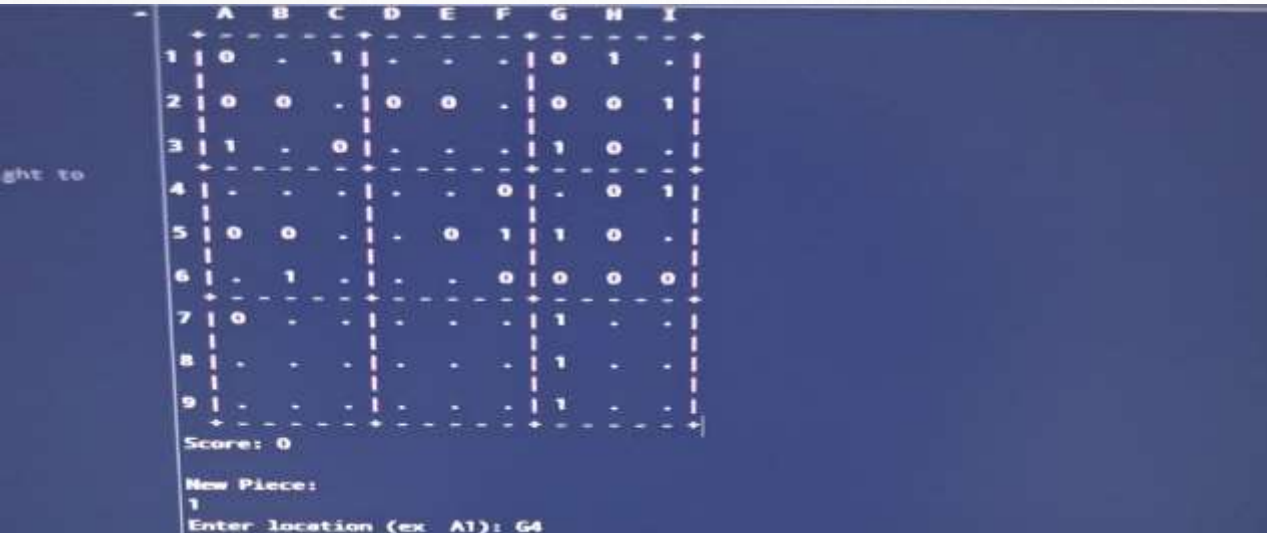
- Game including 10 piece element and they are different from each other.
- Elements are created randomly points that connecting to other points in 3X3 block.
- When player use a elements game picks one of this 10 pieces randomly.

```
void generate_piece(int piece[4][4], int* width, int* height) {  
    int num = rand() % 10;  
    *width = piece_widths[num];  
    *height = piece_heights[num];  
  
    for (int i = 0; i < *height; i++) {  
        for (int j = 0; j < *width; j++) {  
            piece[i][j] = pieces[num][i][j];  
        }  
    }  
}
```

- *Creates random pieces to be used in the game.*
- *Pieces are generated according to the game rules.*
- *Each piece can be defined as an array or matrix.*

CHAPTER 5

(The pictures of Score calculating part of program)



CALCULATING SCORE

- IF THE LOCATED ELEMENT:
 - FILLS A FULL ROW(S) (AND/OR)
 - FILLS A COLUMN(S) COMPLETELY (AND/OR)
 - FULLY FILLS A 3*3 BLOCK(S),

THE FULLY FILLED
ROW(S)/COLUMN(S)/BLOCK(S) ARE
TREATED AS A BINARY NUMBER. THE
DECIMAL EQUIVALENT OF BINARY
NUMBERS ARE CALCULATED.

- IF THE CURRENTLY PLACED GAME ELEMENT
FILLS MORE THAN ONE
ROW(S)/COLUMN(S)/BLOCK(S), THE
DECIMAL EQUIVALENT OF BINARY
NUMBERS ARE ADDED AND THIS SUM IS
MULTIPLIED BY THE NUMBER COMPLETED
PARTS (ROW(S)/COLUMN(S)/BLOCK(S)).
THIS RESULT IS ADDED TO THE SCORE.
- COMPLETED PARTS ARE REMOVED FROM
THE BOARD.

GAME ENDING

- *Determines the condition under which the game should end.*
- *It calculates the score by taking into account all conditions such as winning, losing or completing all moves.*

```
// CheckEmpty
int game_over(int piece[4][4], int width, int height) {
    for (int i = 0; i < SIZE; i++) {
        for (int j = 0; j < SIZE; j++) {
            if (valid_invalid(piece, width, height, i, j)) {
                return 0;
            }
        }
    }
    return 1; //no placement
}
```